GRRSIS: Generalized Referring Remote Sensing Image Segmentation

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***Abstract*—Referring remote sensing image segmentation (RRSIS) is a challenging task that involves segmenting target instances within a top-view image guided by a natural language expression. Existing classic RRSIS methods commonly support target expressions only, i.e., the target described by the expression is present in the image. No-target expressions are excluded. Under this constraint, the model may face significant challenges. For instance, a small error, such as a typographical mistake, could cause a complete failure of the model. To overcome this issue, in this article, we introduce a new benchmark called generalized RRSIS (GRRSIS), which extends classic RRSIS by allowing expressions to refer to no-target objects. Toward this, we construct the first large-scale dataset for GRRSIS, called GRRSIS-D, which includes multitarget, single-target, and no-target expressions. Core challenges in GRRSIS stem from the fact that objects in aerial images often occupy only a small number of pixels, exhibit significant orientation variations, and present varying levels of recognition difficulty. To tackle these challenges, we propose an oriented-aware multiscale network with an adaptive angle sensing module that integrates adaptive rotated convolution and a gating mechanism to capture diverse object orientations while suppressing irrelevant features for more accurate representations. In addition, we introduce a novel online hard case mining loss, which allocates varying levels of attention to foreground and background regions and reshapes the standard loss by downweighting well-segmented examples, effectively addressing the issues caused by low pixel occupancy and uneven sample difficulty. The proposed approach achieves state-of-the-art performance on both the newly introduced GRRSIS and classic RRSIS tasks.**

***Index Terms*—Adaptive multimodal feature fusion (AMFF), generalized referring remote sensing segmentation dataset, generalized referring remote sensing segmentation, online hard case mining loss.**

I. Introduction

REFERRING expression segmentation (RES) [1], [2], [3], [4], [5], [6] is a crucial task that involves identifying an object in an image based on a natural language description and generating a segmentation mask for it. Recently, RES has been extended to the remote sensing field, giving rise to a new task called referring remote sensing image segmentation (RRSIS) [7], [8]. This task offers significant potential for diverse applications, such as urban planning [9], wetland monitoring [10], [11], [12], land cover monitoring [11], [13], [14]. Currently, most of the existing methods follow the RRSIS rules defined in the popular datasets RefSegRS [7] and RRSIS-D [8], and have achieved great progress in recent years.

However, existing RRSIS methods have some strong predefined constraints. For example, classic methods do not account for no-target expressions that do not match any object in the image. This means that the behavior of the existing RRSIS methods is undefined if the target does not exist in the input image. When it comes to practical applications under such constraints, the input expression must match an object in the image; failing which, problems are bound to arise. The experiment results in Section VI-C also can demonstrate this point. In fact, the object described by the expression is present in the image, which is not always guaranteed in practical applications. In real-world scenarios, it is often unclear whether the object mentioned in the natural language description is actually included in the image.

To address this constraint, we propose a new benchmark named generalized RRSIS (GRRSIS), which accommodates expressions indicating any number of target objects. In GRRSIS, the inputs consist of an image and a referring expression, while the output is a mask corresponding to the referring expression, similar to the classic RRSIS task. In contrast to the classic RRSIS task, as illustrated in Fig. 1, GRRSIS also supports no-target expressions that do not refer to any object in the image, such as "baseball field that is approximately 4460 square meters on the northwest side." This new benchmark allows for more flexible expressions and enhances the utility and robustness of RRSIS in practical scenarios. However, the existing dataset for RRSIS does not contain no-target samples. To facilitate research efforts on realistic referring segmentation, we build a new dataset for GRRSIS, called GRRSIS-D. This dataset contains target and no-target expressions with satellite or aerial images. These images span various spatial resolutions, covering areas ranging from 0.06 to 25 km^2.

Moreover, we propose a baseline method called oriented-aware multiscale network (OMSN) tailored to the objectives of the GRRSIS. We identify several main challenges in this task: 1) the significant variations in object orientation make it challenging for the model to learn an intrinsic feature

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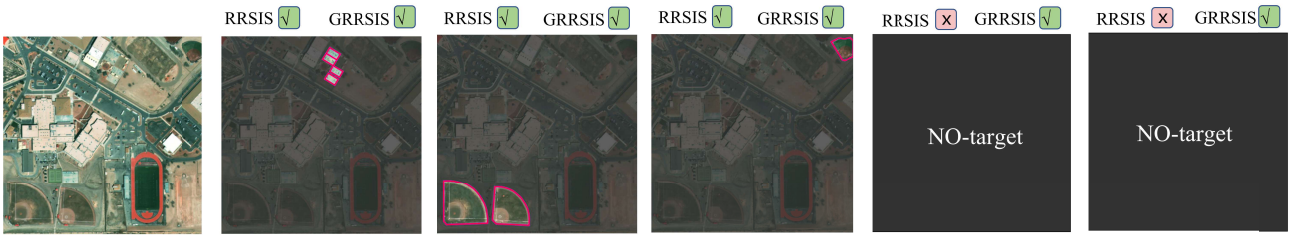
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Image

(a) basketball court on the northeast side

(b) baseball field next to a tennis court

(c) a 4460 square meter baseball field is located in the northeast of a 489 square meter basketball court, about 164 meters away

(d) baseball field that is about 4460 square meters on the northwest side.

(e) a 4460 square meter baseball field is located in the southeast of a 489 square meter basketball court, about 164 meters away

Fig. 1. Classic RRSIS only supports expressions that refer to objects present in the images (a)–(c), whereas the proposed GRRSIS also accommodates expressions that refer to objects not present in the images (d) and (e).

representation; 2) the small fraction of total pixels occupied by targets in remote sensing images leads to a severe foreground–background imbalance, increasing the likelihood of missing detection targets. This issue is further exacerbated by the introduction of no-target sample; and 3) the varying difficulty levels of sample recognition cause the model to focus predominantly on simpler samples, hindering its ability to effectively learn from harder ones.

To tackle challenge 1), our OMSN incorporates an adaptive multimodal feature fusion (AMFF) module with an adaptive angle sensing (AAS) component to adaptively capture the orientations of objects described in expressions. Within the AAS module, a gating mechanism adaptively modulates the orientation-aware features extracted by the adaptive rotated convolution (ARC) before integrating them with language-guided visual features, thereby ensuring a positive contribution to the final representation. To tackle the challenge 2) and 3), a novel online hard example mining loss is designed to guide model training. Specifically, online hard example mining loss allocates varying levels of attention to foreground and background within the model to mitigate the issue of target loss. This approach encourages the model to focus on foreground information while penalizing misclassified pixels, especially when nontarget pixels in no-target expression are incorrectly classified as positive. Furthermore, to mitigate the challenges posed by uneven sample difficulty, the loss function incorporates a dynamically scaled factor. This factor adjusts the weight of each sample during training, with higher weights assigned to harder examples and decaying as confidence in the correct segmentation increases. This dynamic scaling helps ensure that the model is effectively trained on both easy and challenging samples, improving its generalization capabilities.

In summary, our contributions are listed as follows.

1) We propose the GRRSIS benchmark, which extends traditional RRSIS by allowing language expressions that do not correspond to any target in the image, thereby enhancing adaptability and robustness in complex scenarios.
2) We propose an OMSN model for GRRSIS, incorporating a novel AAS module that adaptively captures the orientations of target objects. In addition, we design an online hard example mining loss that emphasizes foreground pixels and progressively reduces attention to well-segmented examples, thereby improving training efficiency.
3) We propose GRRSIS-D, the first benchmark dataset designed for the GRRSIS, along with an automatic method for generating ground-truth masks from natural language expressions. The dataset comprises 106 029 image-language-label triplets and covers a wide variety of object categories with significant scale variations.
4) Extensive experiments show that the proposed OMSN achieves state-of-the-art performance on the GRRSIS benchmark, significantly boosting segmentation accuracy. Furthermore, it also demonstrates strong generalization ability on the traditional RRSIS task.

II. Related Work

A. Related Referring Tasks and Datasets

RRSIS involves identifying objects in remote sensing images and generating segmentation masks based on natural language descriptions [7], [8]. It is an emerging task that extends from RES. Although RES has achieved remarkable success in natural scene images over the past few decades, there is limited research and few datasets focused on RRSIS. Currently, there are two datasets specifically designed for RRSIS: RefSegRS [7] and RRSIS-D [8]. However, these datasets contain only target expressions. Models trained on these datasets are prone to errors when encountering expressions that do not correspond to any object in the image, such as those containing typographical errors. Although the original definition of RRSIS does not restrict the number of target instances, "the instance must be included in the image" has become a de facto rule for the RRSIS task.

In addition to the RRSIS task, another task closely related to our GRRSIS is the generalized RES (GRES) task. GRES [15], an extension of RES, broadens the scope of referring segmentation in natural scenes to include both target expressions and no-target expressions, significantly enhancing its practical application range. Currently, there is one dataset for this task, gRefCOCO [15], which contains a wealth of natural scenes. However, due to the differences between natural and remote sensing images, this dataset cannot be directly applied to remote sensing. Inspired by this task, we extend the RRSIS task to also recognize no-target scenarios, and propose a

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dataset that contains no-target samples, addressing a key gap in existing resources for the GRRSIS.

B. Methods for RES, RRSIS and GRES

Previous work on RES has primarily focused on theory and methodology, with an emphasis on efficiently extracting visual and linguistic features and bridging the heterogeneity between image and language modalities for effective information fusion. High-quality feature extraction is essential for the RES task. Early models typically employed CNNs [16], [17] for visual feature extraction and recurrent neural networks (RNNs) [18], gated recurrent units (GRUs) [19] or long short-term memory networks (LSTMs) [20], [21], [22] for language feature extraction, projecting both modalities into a shared space before performing feature fusion [23], [24], [25]. With the advent of Transformer [26] models, Swin-Transformer [27] has become the dominant backbone for visual feature extraction [28], while bidirectional encoder representations from transformers (BERTs) [29] is widely used for language feature extraction [30]. For feature fusion, early methods relied on simple feature concatenation [24], [31], which has since evolved into various attention mechanisms that enhance the interaction between visual and linguistic features [25], [32], [33]. These approaches further refine the extracted features through post-processing. More recent methods have begun to incorporate language information directly into the visual feature extraction process, enabling more language-relevant feature extraction with promising results [30].

As the field has progressed, researchers have considered the inherent limitations of the RES task and proposed a series of extended tasks, such as RRSIS [7], [8], and GRES [15]. These extended tasks have inspired a range of new theoretical advancements designed specifically to meet their unique demands.

RRSIS methods always build upon the theoretical foundations of RES, with adaptations to meet the unique challenges of remote sensing. For example, language-guided cross-scale enhancement (LGCE) [7], which follows the feature extraction approach of language-aware vision transformer (LAVT), addresses the large-scale variations commonly seen in remote sensing images by leveraging linguistic features to adaptively enhance multiscale visual features. Similarly, the rotated multiscale interaction network [8] builds upon LAVT by further addressing scale variations and the diverse orientations of objects in remote sensing scenes. However, these methods do not account for no-target samples, which often leads to incorrect mask generation when such samples are present. Our tests of these methods within the GRRSIS framework revealed suboptimal performance, as discussed in Section VI-C.

GRES methods, such as ReLA [15], model the complex relationships among objects and have proven effective in handling GRES tasks. Yet, due to the significant differences between natural and remote sensing images, its direct application to GRRSIS is ineffective. We use ReLA as a baseline and develop our model to accommodate the requirements for handling no-target samples while also considering the unique characteristics of remote sensing images.

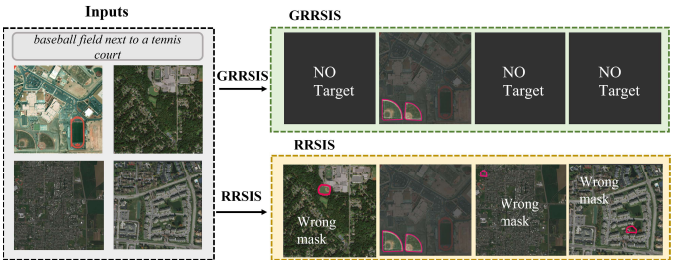


Fig. 2. More applications of GRRSIS brought by supporting no-target expressions compared with classic RRSIS.

III. GRRSIS Setting

A. Revisit of RRSIS

Classic RRSIS takes a remote sensing image and an expression as inputs, aiming to output the mask of the object represented by the expression. As discussed in Section I, current RRSIS models do not consider no-target expressions. Consequently, existing models are likely to incorrectly output an instance if the input expression refers to nothing in the input image.

B. Generalized RRSIS

To address the limitations of classic RRSIS, we propose an expanded task called GRRSIS, which allows for expressions that indicate nothing in the image. A data sample in RRSIS consists of an image I, an expression T related to an object, and a mask M_{GT} corresponding to the referred object. As an extension of the RRSIS task, GRRSIS further introduces a binary label that indicates whether the sample is a no-target expression. The number of instances in T is not restricted. For no-target expressions, M_{GT} should be entirely negative. GRRSIS takes I and T as inputs and predicts both the mask M and the binary label.

The extension of referring segmentation into more realistic scenarios is significant. For example, with the support of no-target expressions, users can apply the same expression to a set of images to identify which images contain the object (s) described by the expression (as Fig. 2 shown). This aligns well with the needs of remote sensing scenarios, where images are typically large and need to be cropped into groups for processing. Users can focus on their queries without worrying about whether the expression used for object retrieval matches the objects in the images. The extension also greatly enhances the model's reliability and robustness in real-world situations, where any type of expression may arise unexpectedly. For example, users providing natural language input may not have access to real-time images.

IV. GRRSIS-D Dataset

A. Dataset Creation

To perform the GRRSIS task, we construct the GRRSIS-D dataset. Images in the RSVG dataset [34] are obtained from multiple sensors and platforms, capturing scenes across various cities in the United States and providing a certain level of contextual richness and representativeness. For convenience, we use these images to directly generate the GRRSIS-D

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dataset. When images are provided, an automated process is designed to capture object instances and generate their corresponding expressions in our work, and the details are as follows.

Step 1 (Referring Expression Template Generation): To efficiently generate grammatically correct referring expressions, we first design a series of language templates with placeholders based on common object description practices in the remote sensing field. These placeholders primarily include object categories, attributes, and spatial relationships between objects. Subsequently, we will fill in these language templates to generate referring expressions. The details and principles of the template design are presented in Section IV-B.

Step 2 (Target Expression and Mask Generation): With the templates prepared, referring expressions are generated by populating the predefined templates. Each image in the RSVG dataset is tied to a central geographic coordinate and image resolution information. Using this data, we map the image area onto OpenStreetMap. From this alignment, we extract ground objects with annotated attributes such as geolocation, boundary, area, shape, and distance. These attributes are then used to fill the placeholders in templates. For example, given the template "< obj >," we can generate the expression "pond" and extract the polygons of all ponds from OpenStreetMap. These polygons are then projected back onto the image plane to generate the ground-truth mask.

Step 3 (No-Target Sample Generation): We introduce category and attribute perturbations into existing expressions to generate potential no-target expressions. For example, for expressions based on the template type "< obj >," we randomly select a category not present in the image as a perturbation, replacing the original category to create a no-target expression. Similarly, for the template type "< obj > on the < po − att >," we can apply category perturbation and absolute position attribute perturbation. After generating the no-target expressions, we compare them with all objects in the image to ensure that the generated no-target expressions are unrelated to the image. In this way, the image, the no-target expression, a zero mask of the same size as the image, and the true label are combined to form the no-target samples.

Step 4 (Post-Processing): Due to potential discrepancies between image collection and OpenStreetMap updates, such as demolished structures, we engage volunteers to manually review the data. Samples that could not be identified based on the referring expressions are removed. This careful curation ensures the dataset's accuracy and finalization.

This method efficiently generates crowdsourced descriptions and annotations for target/no-target expressions in remote sensing images at a low cost. Given that no-target samples are less informative than single- or multitarget samples, we retain at most ten no-target expressions per image. Moreover, to ensure diversity, we strictly limit each template to generate no more than five no-target expressions for the same image.

B. Statistical Analysis

The constructed GRRSIS-D dataset contains a total of 106 029 image-expression-mask triples, including 68 869 target expressions and 37 160 no-target expressions across 4239 images. Specifically, there are 71 879 image-expression-mask triples for training, 17 049 for validation, and 17 110 for testing. This dataset contains 11 categories (as shown in Table I), with each image associated with an average of 25.01 referring expressions and an average expression length of 13.11 words. The statistical information is shown in Table II, and some examples from this dataset are illustrated in Fig. 3. As described in Section I, our GRRSIS-D dataset extends the RRSIS task to scenarios where expressions may be unrelated to the images, significantly enhancing model robustness. In addition, our dataset features unique designs tailored to remote sensing scenarios. We list some unique and significant features of our dataset as follows.

TABLE I
OBJECTS AND ATTRIBUTES IN GRRSIS-D

Feature	abbreviation	label
object	obj	baseball field, basketball court, tennis court, football field, ground track field, swimming pool, storage tank, water tower, bridge, and parking lot.
absolute position attribute	po-att	north, south, east and west, southeast, northeast, southwest, southwest
area attribute	area-att	278, 2560, 171, 3295, 11573, 5199, etc
area threshold	area-th	500, 12000,4000,
relative position attribute	rel-po-att	north, south, east and west, southeast, northeast, southwest, southwest
distance attribute	dis-att	180, 61, 79, 53, 50, 42, 16, 157, 98, etc

TABLE II
STATISTICAL INFORMATION FOR GRRSIS-D

Dataset	referring expressions (target / no-target expression)	image	average expression length
total	106029 (68869/37160)	4239	13.11
train	71870 (45810/26060)	2963	13.16
val	17049 (11524/5525)	636	12.95
test	17110 (11535/5575)	640	13.06

Target Samples: In practice, users often need to find all targets that meet certain conditions. For example, they may want to identify all tennis courts in a remote sensing image or locate tennis courts with specific attributes, such as those next to a swimming pool. At this point, we may not accurately know whether the image contains multiple or single objects, or if it does not include the objects described by the user. Based on this, we propose seven types of language templates, as shown in Table III, guided by the following two principles.

1) Target and Its Own Attributes: Categories and attributes are commonly used to indicate interest in specific objects. Our proposed language templates incorporate both categories and their associated attributes. Given that our goals are identified all objects in the image that match the expression and the unique requirements for geometrical information (such as object orientation, area, and distance) of the remote sensing field, our dataset does not include many relative attributes like "the largest baseball field," which are often used in traditional datasets. Instead, we focus on describing objects

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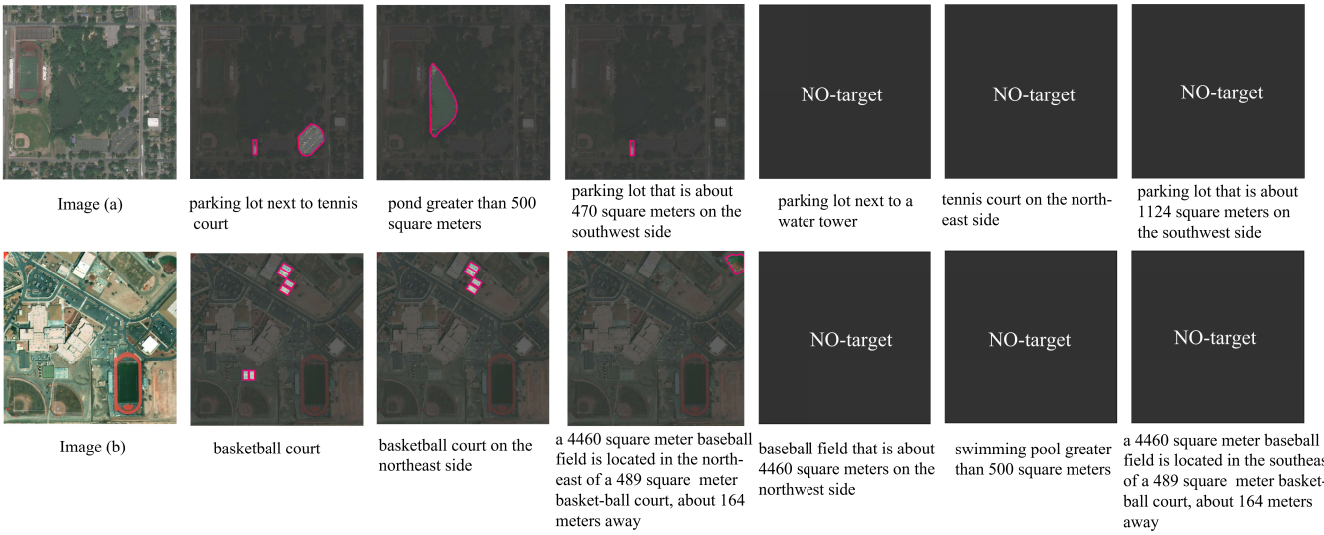


Fig. 3. Examples of the proposed GRRSIS-D dataset.

TABLE III

EXPRESSION TEMPLATES: WHERE < OBJ > REPRESENTS THE OBJECT CATEGORY, AND < PO − ATT >, < AREA − ATT >, < REL − PO − ATT >, < DIS − ATT >, AND < AREA − TH > REPRESENT ATTRIBUTES SUCH AS ABSOLUTE POSITION, AREA, RELATIVE POSITION, DISTANCE, AND AREA THRESHOLD, RESPECTIVELY

Index	Exemplar templates	Expression examples
#1	$< obj >$	pond
#2	$< obj >$ on the $< po-att >$ side	parking lot on the southeast side
#3	$< obj >$ that is about $< area-att >$ square meters on the $< po-att >$ side	pond that is about 948 square meters on the southwest side
#4	$< obj >$ greater than $< area-th >$ square meters	storage tank greater than 500 square meters
#5	a $< area-att >$ square meter $< obj0 >$ is located in the $< rel-po-att >$ of a $< area-att >$ square meter $< obj1 >$, about $< dis-att >$ meters away.	a 2393 square meter parking lot is located in the northeast of a 11135 square meter pond, about 135 meters away
#6	a $< area-att >$ square meter $< obj0 >$ is next to a $< area-att >$ square meter $< obj1 >$, the distance is about $< dis-att >$ meters	a 257 square meter tennis court is next to a 13282 square meter ground track field, the distance is about 15 meter
#7	$< obj0 >$ next to a $< obj1 >$	tennis court next to a parking lot

using absolute position attributes, area attributes, and area threshold attributes, as shown in Table I. Unlike gRefCOCO, our expressions do not feature counting expressions like "the two people on the far left" or compound sentence structures like "A and B." which could allow the model to easily infer the number of target objects based on sentence structure, potentially leading to shortcuts during training. Instead, we aim for more realistic scenarios by categorizing class and attribute information to identify all instances that meet certain conditions, i.e., "pond greater than 500 square meters," regardless of their existence. The templates designed based on this approach include: #1, #2, #3, and #4.

2) Object With a Reference: Describing objects using a reference object is a common technique for creating referring expressions, especially in the remote sensing field. Since remote sensing images are captured from a top–down perspective, we focus on two primary spatial relationships: "next to" and other relevant spatial relationships. The term "next to" indicates that the distance between two objects is less than 50 m, with no significant obstacles between them. In addition, we incorporate relative position and distance attributes to describe the relationships between the target object and its reference. Relying solely on spatial relationships, such as stating that "a swimming pool is to the northeast of a basketball court" can be ambiguous, particularly when the two objects are thousands of kilometers apart, which diminishes the usefulness of such retrieval. Therefore, we introduce distance attributes in our descriptions. This approach generates templates #5, #6, and #7, as shown in Table III.

No-Target Samples: An easy way to obtain no-target expressions is by randomly selecting expressions from the generated sentence pool that do not match any targets in the image. However, we found that if no constraints are imposed on the selection of no-target expressions, these samples tend to be overly simplistic and significantly different from expressions with valid targets. For example, an expression like "pond that is about 110 square meters" might be selected for images that do not contain a pond. To avoid generating an excessive number of simplistic samples that can be easily judged based solely on object categories, we established some rules for generating no-target expressions to create more deceptive representations.

a) Attribute Priority: If no-target expressions can be generated using attribute perturbations, there is no need to use category perturbations. For example, for template #2, we can apply either category perturbations or absolute position attribute perturbations to generate no-target samples. To enhance the deception, we prefer to use attribute perturbations (absolute position attribute) instead of category perturbations when generating no-target expressions.

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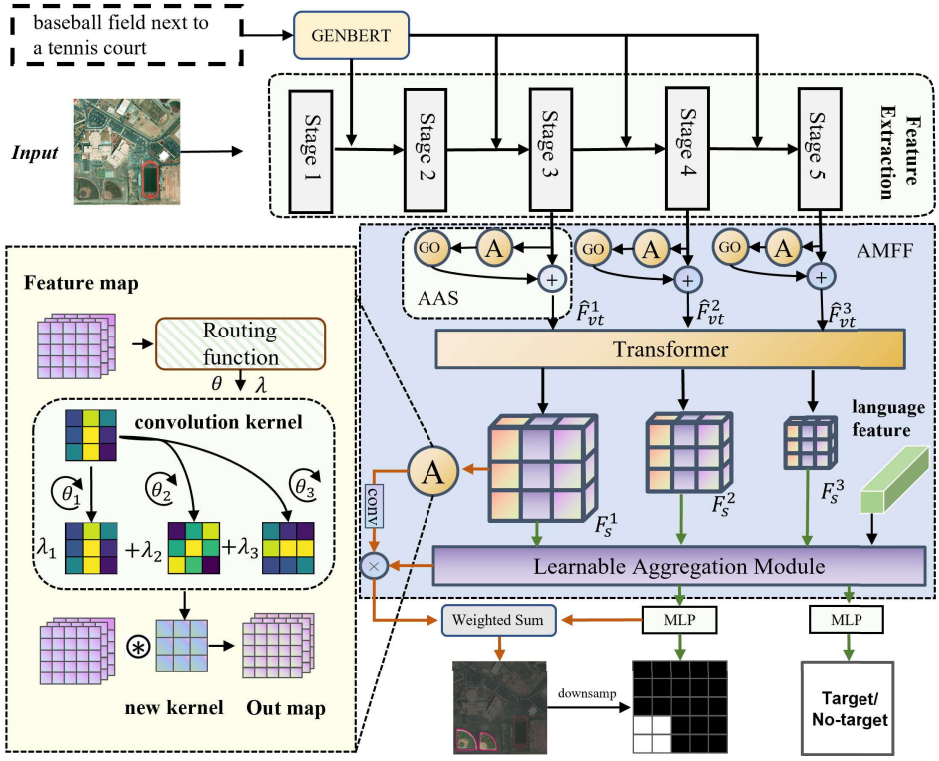


Fig. 4. Overview of the proposed OMSN. Our AMFF module incorporates an AAS component to adaptively capture object orientations described in expressions, along with cross-scale attention to address the scale variations of these objects.

b) Distance Limitations: Distance perturbation is one of the important methods for generating no-target samples, with a range of $(0, +\infty)$. The difficulty of recognizing the generated negative samples in this way varies significantly. Therefore, we impose certain constraints on the distance attributes, specifying that the difference between the perturbed distance and the original distance must be between 20 and 200 m. In cases with "next to" relationships, the upper bound perturbation distance is 50 m. The minimum difference between the perturbation distance and the original distance is set to 20 m to ensure the expressions are challenging yet recognizable. These rules greatly improve the diversity of no-target expressions and keep our dataset at a reasonable difficulty.

These rules greatly improve the diversity of no-target expressions and keep our dataset at a reasonable difficulty.

V. Approach

As discussed earlier, the significant variations in object orientation within remote sensing imagery make it challenging to learn an intrinsic feature representation. Our OMSN incorporates an AMFF module with an AAS subcomponent to handle diverse object orientations. In addition, to effectively handle scale variations, AMFF leverages a deformable transformer for cross-scale interaction and a Learnable Aggregation Module [15] for multiscale feature fusion. The framework of OMSN is shown in Fig. 4. Moreover, we observe that the small scale of target objects and the wide variation in recognition difficulty hinder effective model learning. To address this issue, we propose an online hard example mining Loss to guide the training of the model.

A. Model Architecture

1) Language Encoder: Considering that in remote sensing applications, attributes like object area and the distance between objects are often presented numerically in expressions such as "basketball court that is about 1253 square meters on the southeast side." These expressions require our language feature extraction model to understand numbers and perform numerical reasoning. For example, descriptions like "pond greater than 500 square meters" also demand numerical reasoning. Therefore, we utilize the GENBERT [35] model, a simple BERT-based generative model that performs numerical computations internally, for language feature extraction. The GENBERT model is pretrained on synthetic numerical question-answer pairs, enabling it to perform numerical reasoning.

To enhance the understanding of numerical information, we employ a more efficient digit-by-digit approach for number embeddings. For example, a wordpiece represented as $\#\#d_1 \cdots d_k$, where $d_i \in \{0, \cdots, 9\}$, is further split into $\#\#d_1, \#\#d_2, \cdots, \#\#d_k$. The embedding method for other words remains the same as in the traditional BERT model.

2) AMFF Module: Objects in remote sensing images appear at various angles, and traditional static convolution kernels are unable to effectively extract their features. This is one of the main reasons why mask generation may result in a loss of precision [8]. Currently, many practitioners working with remote sensing images have recognized this issue and proposed ARC strategies for object detection tasks [36], [37], [38]. Inspired by these works, we introduce ARC into our AMFF module to achieve better mask prediction.

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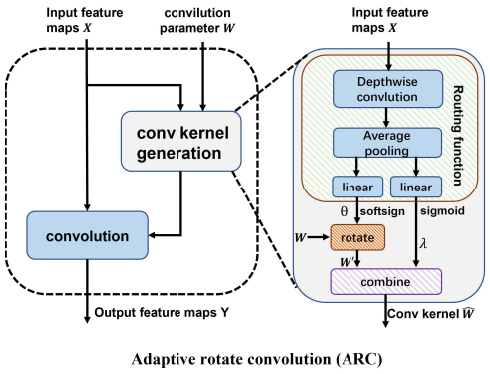


Fig. 5. Structure of ARC.

ARC [36] captures image features from various angles by dynamically adjusting the convolution kernels, as shown in Fig. 5. Specifically, it adopts a shallow network to jointly predict the convolution kernel angle $\boldsymbol{\theta}$ and the feature aggregation weights $\boldsymbol{\lambda}$. The feature map is then adjusted using $\boldsymbol{\theta}$ and $\boldsymbol{\lambda}$ to better adapt to objects at various angles. Specifically, given an input feature X, $\boldsymbol{\theta}$, and $\boldsymbol{\lambda}$ are predicted as follows:

$$\boldsymbol{\theta}, \boldsymbol{\lambda} = \text{Routing}\,(X)\,. \tag{1}$$

More precisely, the structure of Routing(·) is illustrated in Fig. 5, consisting of a depthwise convolution layer, an average pooling layer, and two linear layers. Here, $\boldsymbol{\theta} = [\theta_1, \theta_2, \cdots, \theta_n]$ and $\boldsymbol{\lambda} = [\lambda_1, \lambda_2, \cdots, \lambda_n]$. Each convolution kernel W_i is rotated by θ_i, and the convolved features are then aggregated using $\boldsymbol{\lambda}$ as follows:

$$Y = \sum_{i=1}^{n} \lambda_i \text{Rotate}\left(W_i^1; \theta_i\right) * X \tag{2}$$

where Rotate(·) is the rotation function, which means that the new kernel is obtained by rotating the original coordinates clockwise around the center point by θ_i degrees. An intuitive explanation is that convolution kernels at different angles are applied to the input features, which are then weighted and aggregated according to their importance. This strategy enhances the network's representational capacity for capturing features of objects at multiple orientations.

Ensuring that the generated rotation angles and weights are more focused on the object described by the language is key to improving the model's effectiveness for GRRSIS. To achieve this, the language-aware visual encoder from LAVT [30] is used in our OMSN to extract language-related visual features. These language-related visual features, extracted at resolutions of 1/4, 1/8, 1/16, and 1/32 of the original image, are denoted as F_{vt}^i, where $i \in 0, 1, 2, 3$.

To adaptively perceive the features of rotating objects, we design an AAS module based on ARC. We observe that the features produced by ARC exhibit more concentrated attention, and the introduction of ARC enables high-quality features to better adapt to various object orientations. However, applying ARC to low-quality features can negatively impact the results, as demonstrated in Section IV. Therefore, to ensure that the ARC module contributes positively to performance, we incorporate a gating mechanism for incremental control.

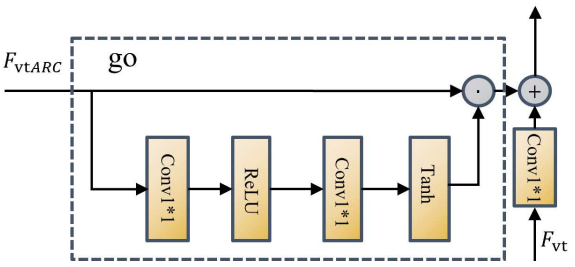


Fig. 6. Gate mechanism.

Taking the processing of F_{vt}^1 as an example, the AAS operates as follows, with F_{vt}^2 and F_{vt}^3 undergoing similar processing:

$$F_{\text{vtARC}}^1 = \text{ARC}\left(F_{\text{vt}}^1\right) \tag{3}$$

$$\hat{F}_{\text{vt}}^1 = \text{conv}\left(F_{\text{vt}}^1\right) + \text{go}\left(F_{\text{vtARC}}^1\right) \tag{4}$$

where go(·) represents the gate. This operation serves to retain part of the original features while selectively introducing beneficial components from the ARC-enhanced features. By doing so, it ensures that incorporating the ARC module does not negatively affect the experimental results. Furthermore, when the ARC module successfully extracts sufficiently informative features, the contribution of the original features becomes less significant and can be considered negligible. The structure of the go(·) module is illustrated in Fig. 6. Specifically, the gate consists of two sequential layers: the first is a 1×1 convolution followed by a rectified linear unit (ReLU) activation, and the second is another 1×1 convolution followed by a hyperbolic tangent (tanh) activation function.

Following the AAS module, to more fully integrate information from features at various scales, we use a deformable transformer (deform-trans) [39] to enhance $\hat{F}_{\text{vt}}^i$ for $i \in 1, 2, 3$. In practice, we first flatten the feature maps and then process them through the transformer layer to obtain fused features F_s^i for $i \in 1, 2, 3$, as described below

$$\left[F_s^1, F_s^2, F_s^3\right] = \text{Tra}\left(\left[\hat{F}_{\text{vt}}^1, \hat{F}_{\text{vt}}^2, \hat{F}_{\text{vt}}^3\right]\right) \tag{5}$$

where $[\cdot]$ denotes concatenation, and Tra(·) represents the deformable transformer. The resulting scale-enhanced features F_s^i effectively integrate the advantages of features from multiple scales.

Considering that remote sensing images are typically large in scale, analyzing them in segmented regions is more effective for accurately locating objects described by text. Thus, in addition to the pixel-level and instance-level prediction branches, our model incorporates a region-level prediction branch. The instance-level prediction branch aims to determine the absence of target objects, while the region-level prediction branch determines whether small regions contain pixels corresponding to the language-described object. The pixel-level prediction branch performs fine-grained pixel classification. Following previous work [15], we use $P \times P$ learnable regions Q to capture information from F_s^1, F_s^2, F_s^3 and text L based on the attention mechanism, and build a learnable aggregation module as follows:

$$F_{\text{qv}} = \text{self}\left(\left(Q^T F_s^i\right) \cdot F_s^i\right) \tag{6}$$

$$Q = F_{\text{qv}} + Q \tag{7}$$

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$$F_{\text{qvt}} = \text{self}\left(\left(Q^T f(L)\right) \cdot f(L)\right) \tag{8}$$

$$Q = F_{\text{qvt}} + Q \tag{9}$$

where self(·) denotes the self-attention operation. The obtained region features $F_m = Q = \{f_r^n\}_{n=1}^{P^2}$ are then processed via global pooling and fed into a multilayer perceptron (MLP) to predict whether the sample is a no-target sample, as follows:

$$I = \text{MLP}\left(\text{globalpool}\left(F_m\right)\right) \tag{10}$$

and F_m is also used to predict the probability x_r^n of a target being present in each region

$$x_r^n = \text{MLP}\left(f_r^n\right). \tag{11}$$

In the pixel-level prediction branch, F_{vt}^1 is processed by ARC to ensure that the generated rotation angles and weights are more focused on the objects described by the language. In addition, we use x_r^n to refine the final fused features, which helps produce more accurate segmentation results. The final predicted mask is obtained by a weighted aggregation of these masks

$$\begin{aligned} M_r &= f_r \text{conv}_{1\times 1}\left(\text{ARC}\left(F_{\text{vt}}^0 + \text{upsample}\left(F_s^1\right)\right)\right) \\ M &= \sum_n \left(x_r^n M_r^n\right) \end{aligned} \tag{12}$$

where upsample is the upsampling operation, and F_{vt}^0 is used to fill in the details.

B. Online Hard Case Mining Loss

In the GRRSIS task, target objects are typically small, dense, and exhibit large-scale variations. The ground-truth masks for these small targets are highly imbalanced between foreground and background pixels. The presence of no-target samples intensifies the problem. Training the model with the standard cross-entropy loss leads to a higher tendency to classify pixels as background rather than foreground, exacerbating the issue of missed targets. In addition, the varying spatial resolutions of remote sensing images, significant scale changes of target objects, and their clustered or scattered distributions lead to considerable differences in object recognition difficulty. These factors, in turn, pose significant challenges for model training. Due to the imbalance between easy and hard samples, the model tends to favor recognizing simpler instances while neglecting more challenging ones, resulting in suboptimal performance.

To address the aforementioned issues, the loss function should meet at least three requirements to effectively guide model training: 1) it should place greater emphasis on foreground regions than background, ensuring that object scale differences do not cause significant fluctuations in loss values; 2) pixels where the predicted foreground targets are completely unrelated to the ground truth should be penalized, particularly in no-target samples, as mislabeling these pixels as positive could lead to inconsistencies between the classification and segmentation branch results; and 3) the model should adjust the loss weights based on the segmentation difficulty of the samples. During training, it should focus more on challenging samples to improve learning effectiveness. Our online hard case mining loss is specifically designed to meet these requirements.

1) Balanced Cross Entropy: The loss function with coefficient α_1 is a commonly used method to address class imbalance. However, it remains limited in effectively mitigating the issue of target loss caused by the severe imbalance between foreground and background pixels. Building on previous work [15], which employed a multilevel loss incorporating α_1 and achieved promising results, we adopt this multilevel loss as our baseline. In multilevel loss, the predicted mask M is supervised by the ground-truth target mask M_{GT}

$$\begin{aligned} L_{\text{pixel}} = -\frac{1}{N}\sum_{i=1}^{N}\Big[&\alpha_1 y_p^i \log\left(p^i\right) \\ &+(1-\alpha_1)\left(1-y_p^i\right)\log\left(1-p^i\right)\Big] \end{aligned} \tag{13}$$

where α_1 is a weighting factor used to address the class imbalance problem. y_p^i and p^i are the ground-truth value and the predicted value of pixel i, respectively, and N is the total number of pixels. The $P \times P$ probability map x_r is supervised by a "minimap," which is a 10×10 version downsampled from M_{GT}

$$\begin{aligned} L_{\text{minmap}} = -\frac{1}{P^2}\sum_{i=1}^{P^2}\Big[&\alpha_1 y_r^i \log\left(x_r^i\right) \\ &+(1-\alpha_1)\left(1-y_r^i\right)\log\left(1-x_r^i\right)\Big] \end{aligned} \tag{14}$$

y_r^i and r^i are the ground truth and predicted values of region i. The global average of all region features is used to predict the no-target label

$$\begin{aligned} L_{\text{image}} = -\Big[&(1-\alpha_1)\, y_I \log(I) \\ &+\alpha_1 (1-y_I)\log(1-I)\Big]. \end{aligned} \tag{15}$$

The total loss function is formulated as $L_{\text{loss}} = L_{\text{pixel}} + L_{\text{minmap}} + L_{\text{image}}$.

2) Online Hard Case Mining Loss: To address the aforementioned requirements (1) and (2), we introduce an improved Dice loss alongside the original cross-entropy loss. Unlike cross-entropy, the improved Dice loss emphasizes the similarity between the predicted foreground pixel set and the ground-truth foreground pixel set, and is defined as follows:

$$L_{\text{diceim}} = 1 - \frac{2|X \cap Y| + \varepsilon}{|X| + |Y| + \varepsilon} \tag{16}$$

where X and Y represent the predicted foreground set and the ground-truth foreground set, respectively, and $|\cdot|$ denotes the area. We introduce a small positive constant ε into the original Dice loss. In the case of no-target samples, when the prediction is correct, both $|X \cap Y|$ and $|X| + |Y|$ are zero. Without ε, the original Dice loss becomes undefined, making it unsuitable for the GRRSIS task.

L_{diceim} places greater emphasis on capturing foreground pixels, effectively addressing the foreground–background imbalance problem. For no-target samples, where $|X \cap Y| = 0$, if any pixels are predicted as positive, $|X| + |Y|$ will not be zero, resulting in a large loss that effectively penalizes falsely predicted positive pixels. Incorporating the improved Dice loss into the segmentation branch's loss function can

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significantly mitigate the inconsistency between the results of the classification and segmentation branches. Therefore, the combined loss $L_{\text{pixelcombine}} = L_{\text{pixel}} + L_{\text{diceim}}$ is proposed to enable pixel-level supervision.

More formally, to address the aforementioned requirements (3), inspired by focal loss, a modulating factor is designed to be added to the combined loss $L_{\text{pixelcombine}}$. The modulating function is defined as follows:

$$\delta = \begin{cases} 1, & \text{imou} < \text{imou}_{\text{th}} \\ (1 - \text{imou})^{\gamma}, & \text{imou} >= \text{imou}_{\text{th}} \end{cases} \tag{17}$$

where tunable focusing parameter $\gamma \geq 0$, and $\text{imou} = \frac{|X \cap Y| + \varepsilon}{|X| + |Y| + \varepsilon}$. Our loss for pixel classification can be replaced by

$$L_{\text{pixelhard}} = L_{\text{pixel}} + \delta L_{\text{diceim}} \tag{18}$$

where for target samples, imou $\approx$ IoU reflects the degree of overlap between the predicted mask and the ground-truth mask. When imou exceeds a predefined threshold imou_{th}, the model is considered proficient at handling this sample, and the weight of that sample is reduced in subsequent training. The degree of reduction is closely related to how well the model performs on the sample. For no-target samples, if the prediction is correct, the resulting imou approaches 1, and similarly, the weight of that training sample is reduced in the next training phase. The loss $L_{\text{pixelhard}}$ dynamically adjusts sample weights throughout training, enabling the model to focus more on challenging samples. This strategy encourages the model to achieve more robust segmentation performance. Accordingly, the total loss is defined as

$$L_{\text{total}} = L_{\text{pixelhard}} + L_{\text{minmap}} + L_{\text{image}}. \tag{19}$$

VI. Experiments

To evaluate the effectiveness of the proposed OMSN model and its associated loss function, we conduct a series of experiments. The implementation details and corresponding results are presented below.

A. Implementation Details

1) Experiment Settings: In our experiments, the visual backbone is pretrained on ImageNet22K [40]; GENBERT is pretrained on numerically enriched synthetic textual data. The model is trained for 150 000 iterations using the AdamW optimizer [41]. During training, a MultiStepLR scheduler is employed to adjust the learning rate, which is initialized at 1×10^{-5} and reduced to 0.1 times its current value at the 110 000th and 140 000th iterations. Throughout the training process, both the visual backbone and the language encoder are not kept frozen but are continuously fine-tuned. To reduce computational cost, the input images are resized to 512×512 during both training and testing.

2) Evaluation Metrics: Following previous work [15], we use Intersection-over-Union (cIoU), generalized Intersection-over-Union (gIoU), Precision@X (P@X), no-target accuracy (N-acc), and target accuracy (T-acc) to evaluate the model performance.

1) *cIoU:* cIoU is an important metric for evaluating overall segmentation performance. cIoU measures the ratio of intersecting pixels to the union of predicted and ground-truth pixels across the entire dataset, reflecting the model's overall ability to capture pixel-level accuracy for textually described objects.
2) *gIoU:* gIoU, which extends the standard mean IoU to include no-target samples, computes the average per-image IoU over the entire dataset. For no-target samples, a prediction is regarded as correct if the model outputs True and the predicted mask contains no foreground pixels. For convenience, such correct predictions are counted as true positives (TP), while incorrect predictions are counted as false negatives (FN). When computing IoU, the IoU value of TP no-target cases is assigned as 1, whereas that of FN cases is assigned as 0.
3) *P@X:* P@X is a metric widely used in multiple segmentation tasks. In each sample, IoU is calculated as the intersection of the predicted mask and the ground-truth mask over their union. P@X calculates the percentage of samples with an IoU higher than threshold X, reflecting the model's ability to accurately localize target objects.
4) *N-acc and T-acc:* N-acc evaluates the model's performance in identifying no-target samples. N-acc is defined as:
$$\texttt{N-acc} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$
which represents the proportion of correct predictions among all no-target samples. A higher N-acc suggests that the model produces fewer false negatives. T-acc, defined as the proportion of correctly predicted target samples among all target samples, is presented as follows:
$$\texttt{T-acc} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{21}$$
where true negative (TN) denotes the correctly predicted target samples, while false positive (FP) denotes the incorrectly predicted ones. A higher T-acc value indicates a lower proportion of FP samples.

B. Ablation Study

1) Necessity for Supporting No-Target Expressions: To demonstrate the necessity and effectiveness of supporting no-target expressions, we compare the performance of the same model (our full model and a variant without the classification branch) trained on: 1) only the target expressions in GRRSIS-D and 2) the complete GRRSIS-D dataset (including no-target expressions). As shown in Fig. 7, the first column showcases samples where the described objects are not present in the image, though the referring expressions are syntactically complete. The model trained on the full GRRSIS-D correctly identifies the absence of the target and produces an appropriate null mask, whereas the model trained solely on target samples generates meaningless outputs. The second column presents simulated common input errors in referring expressions: 1) Truncation error: the original expression—"a 3081 square meters baseball field is located in the southeast of a 404 square

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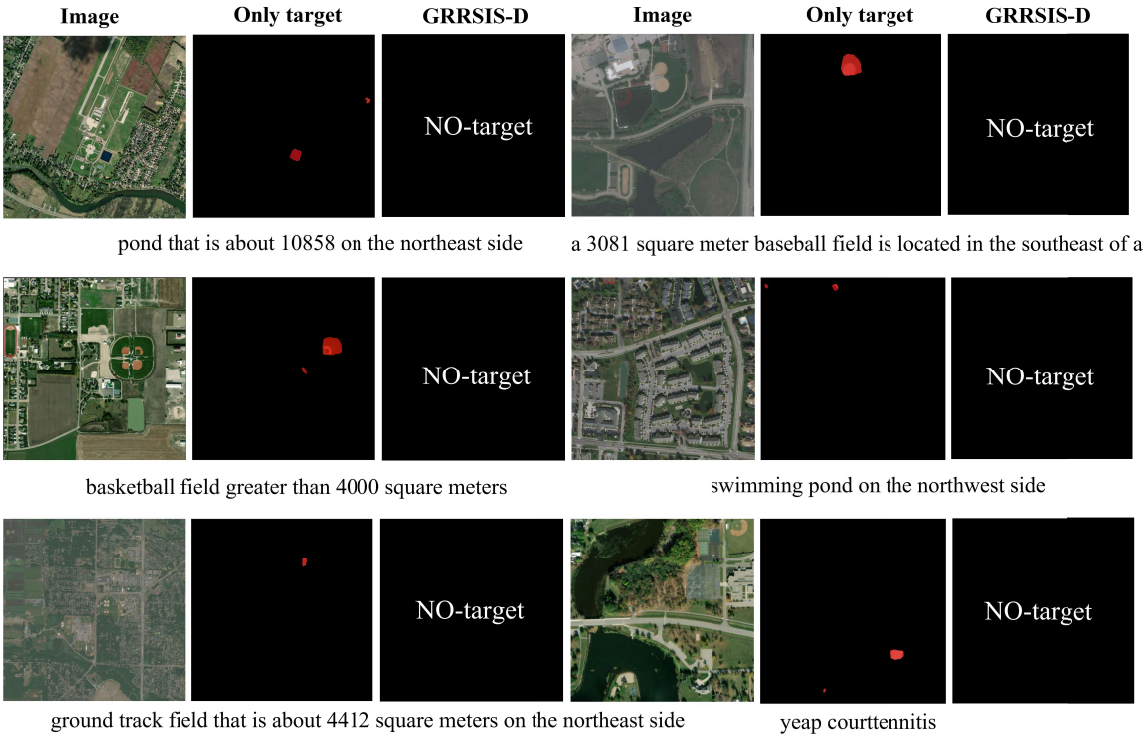


Fig. 7. Example predictions of the same model being trained on a dataset with only target samples versus GRRSIS-D.

TABLE IV
ABLATION STUDY OF MODEL

	baseline	deform-trans	GENBERT	AAS	P@0.5	P@0.7	P@0.9	gIoU	cIoU	T-acc	N-acc
#1	✓				20.89	10.25	0.54	35.58	36.21	83.05	74.85
#2	✓	✓			22.10	11.15	0.97	39.51	39.74	82.41	75.84
#3	✓	✓	✓		23.06	11.91	1.10	39.88	40.34	82.36	75.69
#4	✓	✓	✓	✓	26.15	13.35	0.99	42.58	42.55	80.81	78.28

meters swimming pool, about 105 m away"—is truncated to "a 3081 square meters baseball field is located in the southeast of a;" 2) Typographical error: "swimming pool" in expression is misspelled as "swimming pond;" and 3) Word-order error: the expression is scrambled, rendering it incompatible with any object in the image. In all these cases, the model trained on the full GRRSIS-D is able to correctly generate an appropriate null mask, while the target-only model consistently fails, producing nonsensical masks. These results highlight a critical limitation of models trained exclusively on target expressions: they lack robustness to real-world variability such as no-target queries or input noise. Supporting no-target expressions is therefore essential for practical deployment, ensuring reliable performance across diverse and flexible scenarios.

2) Effictiveness of AAS, Deform-Trans, and GENBERT: In this section, we evaluate the effectiveness of the AAS, deform-trans, and GENBERT on the GRRSIS-D validation dataset through ablation studies. For all experiments, the loss function used is $L_{\text{pixel}} + L_{\text{minmap}} + L_{\text{image}}$. The results are summarized in Table IV. The baseline model is our proposed framework with the AAS and deform-trans modules removed from AMFF, and it adopts the standard BERT model for text feature extraction. In model #2, the deform-trans is introduced into the baseline architecture. Experimental results demonstrate that incorporating deform-trans for cross-scale interaction leads to notable improvements in P@0.5, P@0.7, P@0.9, gIoU, cIoU, and N-acc, while the model's ability to distinguish target samples remains comparable to that of the baseline. In model #3, the standard BERT module is replaced with the proposed GENBERT module, which exhibits greater sensitivity to numerical information. As shown, incorporating GENBERT also yields overall performance gains. In Model #4, the AAS module is integrated into the model. The results indicate that the AAS component in AMFF enhances the model's ability to detect targets with arbitrary orientations, resulting in consistent improvements in P@0.5, P@0.7, and P@0.9 and N-acc.

3) Effictiveness of Online Hard Case Mining Loss: The following experiment evaluates the impact of different components of the online hard case mining loss. The results are shown in Table V. In this experiment, AAS, deform-trans, and GENBERT are included. Model #1 uses $L_{\text{pixel}}+L_{\text{minmap}}+L_{\text{image}}$

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TABLE V
ABLATION STUDY OF LOSS FUNCTION

loss	P@0.5	P@0.7	P@0.9	gIoU	cIoU	T-acc	N-acc
#1	26.15	13.35	0.99	42.58	42.55	80.81	78.28
#2	30.68	15.42	1.17	45.75	42.59	78.85	79.49
#3	31.97	16.83	1.61	47.41	42.63	74.45	83.42

TABLE VI
COMPARISON OF DIFFERENT APPROACH FOR INTRODUCING ARC

	P@0.5	P@0.7	P@0.9	gIoU	cIoU	T-acc	N-acc
w/o ARC	23.06	11.91	1.0	39.88	40.34	82.36	75.69
Approach 1	25.98	13.62	1.17	42.13	42.54	81.93	77.39
Approach 2	24.71	13.21	1.30	40.76	41.70	82.09	75.60
Approach 3	25.39	13.39	1.11	41.69	41.55	81.05	77.86
Approach 4 (AAS)	26.15	13.85	0.99	42.58	42.55	80.81	78.28

as the loss function; model #2 adopts $L_{\text{pixelcombine}} + L_{\text{minmap}} + L_{\text{image}}$; and model #3 adopts $L_{\text{hard}} + L_{\text{minmap}} + L_{\text{image}}$. Comparing the performance of models #1 and #2, we observe that the introduction of $L_{\text{pixelcombine}}$ significantly enhances overall model performance, improving both target sample segmentation and no-target sample segmentation. This supports our hypothesis that emphasizing the foreground in the loss function improves target segmentation accuracy, while increasing the penalty for misclassified samples significantly boosts no-target recognition. Model #3, which uses our online hard case mining loss, further improves performance in gIoU, cIoU, and P@0.5–P@0.9. This suggests that the model gradually focuses on more difficult samples, which is especially effective for GRRSIS tasks.

4) Design Options of AAS: To determine the optimal design of the AAS module, we conduct a series of comparative experiments to evaluate the impact of different ARC integration strategies on model performance. In these experiments, GENBERT is employed for text feature extraction, and the loss function $L_{\text{pixel}} + L_{\text{minmap}} + L_{\text{image}}$ is used to guide model training.

We present four approaches for integrating ARC into our model. In Approach 1, ARC is applied solely to the low-level visual features $F_{\text{vt}}^0 + \text{upsample}(F_s^1)$ in the pixel-level prediction branch. In Approach 2, the obtained multilevel features F_{vt}^1, F_{vt}^2, and F_{vt}^3 are processed through ARC, which are then fused using a transformer-based multiscale feature fusion module. The fused features are utilized in both the region/instance-level and pixel-level prediction branches. Approach 3 is a combination of Approaches 1 and 2. In this method, F_{vt}^1, F_{vt}^2, F_{vt}^3, and $F_c^0 + \text{upsample}(F_c^1)$ are sent to ARC. Approach 4 corresponds to our proposed method AAS, described in Section V-A. The results of these approaches are shown in Table VI.

As shown in Table VI, comparative analysis of the results of Approaches 1–3 with our baseline model (without ARC) reveals that integrating ARC into the multiscale feature extraction process can enhance segmentation performance. However, detailed comparison among the results of Approaches 1–3 suggests that applying ARC to certain feature levels may enhance object segmentation, whereas its application to other levels might adversely affect the results. To further validate this hypothesis, the multiscale features before and after applying rotational convolution in Approach 3 are visualized. The visualization results are presented in Fig. 8. As shown, rotational convolution enables attention to focus on specific bands. For high-quality features (e.g., shallow features in the figure), these bands consistently cover the regions corresponding to the linguistically described objects. Conversely, for lower quality features that were less effective prior to the introduction of rotational convolution (e.g., deep features in the figure), the addition of rotational convolution causes the attention to become even less focused on the object described by the language.

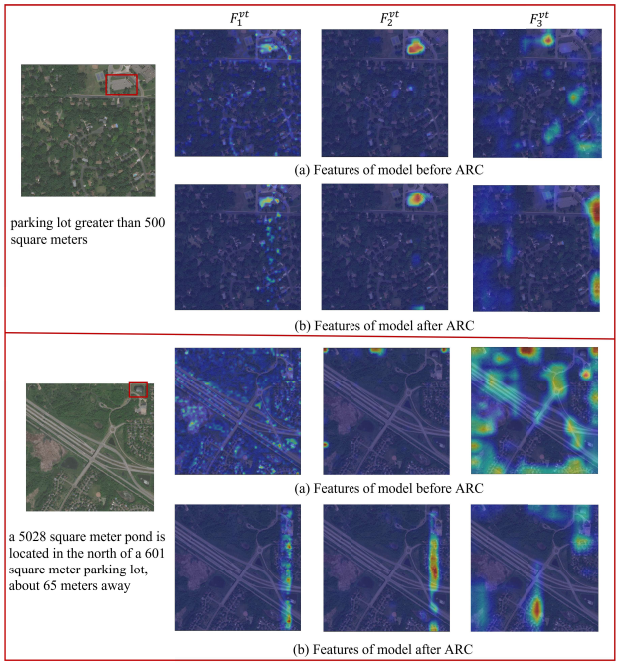


Fig. 8. Feature visualization results.

TABLE VII
MODEL PERFORMANCE WITH DIFFERENT n

	P@0.5	P@0.7	P@0.9	gIoU	cIoU	T-acc	N-acc
n=1	24.44	12.48	0.89	40.86	41.10	82.42	76.29
n=2	13.70	6.15	0.25	32.42	32.60	83.84	69.81
n=4	25.90	13.35	1.09	41.62	41.88	81.67	76.60
n=6	23.74	11.68	1.04	41.62	40.41	81.97	76.07

Based on the above findings, we introduce a gating mechanism into our AAS module to adaptively modulate feature information. This design enables the model to retain beneficial information while suppressing detrimental ones that could impair performance. The experimental results of approach 4, compared with those of approach 3, clearly demonstrate the advantages of incorporating the gating mechanism. Although the numerical improvement may not appear substantial, the introduction of the gating mechanism enables approach 4 to outperform approach 1, indicating that our design effectively filters out detrimental information while preserving useful content.

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TABLE VIII
COMPARISON OF RESULTS FOR OVERALL SAMPLES ON THE GRRSIS-D DATASET

Methods	Visual Encoder	Language Encoder	val				test			
			gIoU	cIoU	N-acc	T-acc	gIoU	cIoU	N-acc	T-acc
CMSA [42]	ResNet-101	None	36.33	23.32	90.73	36.74	34.26	22.04	91.52	34.91
LSCM [43]	ResNet-101	LSTM	37.91	26.89	92.45	36.89	37.13	24.05	92.75	35.22
BRINet [2]	ResNet-101	LSTM	36.14	20.5	**95.16**	26.29	35.87	18.91	**95.70**	23.81
CMPC+ [44]	ResNet-101	LSTM	37.49	24.79	93.35	36.00	37.13	23.24	94.10	34.09
ReSTR [4]	ViT-B-16	Transformer	39.47	34.61	89.65	46.88	40.47	31.96	90.84	46.26
LAVT [30]	Swin-B	BERT	45.42	41.4	86.08	58.12	45.57	39.57	86.48	58.49
CrossVLT [45]	Swin-B	BERT	39.54	31.52	84.62	51.19	40.47	29.68	87.36	50.45
CMIRNet [46]	Swin-B	BERT	45.16	41.04	86.18	55.75	46.14	39.81	88.40	56.25
RMSIN [8]	Swin-B	BERT	45.33	39.27	90.66	52.10	45.78	37.94	90.75	54.24
LGCE [7]	Swin-B	BERT	44.60	41.17	86.00	57.56	45.21	38.34	87.69	57.33
ReLA [15]	Swin-B	BERT	39.51	39.74	75.84	**82.41**	40.12	37.75	77.54	**84.23**
MABP [47]	Swin-B	BERT	40.61	35.75	78.45	58.45	41.04	33.56	79.88	59.83
ours	Swin-B	GENBERT	**47.41**	**42.63**	83.42	74.45	**47.75**	**40.59**	84.09	76.64

TABLE IX
COMPARISON OF RESULTS FOR TARGET SAMPLES ON THE GRRSIS-D DATASET

Methods	val					test				
	P@0.5	P@0.6	P@0.7	P@0.8	P@0.9	P@0.5	p@0.6	P@0.7	P@0.8	P@0.9
CSMA [42]	9.27	6.67	4.35	1.77	0.21	7.64	5.53	3.69	1.55	0.15
LSCM [43]	11.01	8.37	5.54	2.59	0.03	8.76	6.47	4.26	2.04	0.03
BRINet [2]	7.32	5.42	3.53	1.61	0.23	6.10	4.68	3.39	1.59	0.23
CMPC+ [44]	9.85	7.30	4.89	2.13	0.23	8.72	6.24	3.89	1.69	0.26
ReSTR [4]	14.33	9.50	5.49	1.82	0.27	12.96	8.61	4.96	1.84	0.17
LAVT [30]	27.01	20.93	15.42	8.69	2.29	26.74	20.62	15.04	8.75	2.30
CrossVLT[45]	17.24	12.91	9.24	4.96	0.97	17.08	12.71	8.67	4.46	1.04
CMIRNet [46]	27.01	21.98	16.58	**9.39**	**3.13**	27.06	21.75	16.04	**9.70**	**2.80**
RMSIN [8]	24.49	19.45	14.18	8.68	2.61	24.82	19.33	14.07	8.52	2.68
LGCE [7]	25.30	19.69	14.08	8.07	1.70	25.42	19.27	13.39	7.66	1.81
ReLA [15]	22.10	16.33	11.15	5.57	0.97	22.09	16.24	10.62	5.30	0.72
MABP [47]	21.83	16.56	11.68	6.77	1.30	21.79	16.39	10.98	5.90	1.47
Ours	**31.97**	**24.78**	**16.83**	8.71	1.61	**31.18**	**23.96**	**16.19**	8.28	1.49

In addition, the number of rotation angles, n, is an important parameter in AAS. We also evaluate the impact of varying n values on model performance. BERT is used as the language feature extractor, and $L_{\text{pixel}} + L_{\text{minmap}} + L_{\text{image}}$ is employed as the loss function. The experimental results are shown in Table VII. It is evident that as n increases, the model's ability to capture targets initially declines and then improves. The decoder achieves its optimal performance at $n = 4$, where the gIoU is higher than in any other case tested. Therefore, we set $n = 4$ as the default configuration for the ARC module.

5) Design Options of Online Hard Case Mining Loss: In our online hard case mining loss, γ and miou_{th} are two important parameters. In this section, we evaluate the impact of these two parameters on the model's performance. Fig. 9 shows the best gIoU achieved by the model on the validation dataset as γ varies from 0 to 3. When $\gamma = 0$, the loss function reduces to $L_{\text{pixelcombine}} + L_{\text{minmap}} + L_{\text{image}}$. The best performance is achieved when $\gamma = 1$, so we set $\gamma = 1$ as the default value in the online hard case mining loss. In addition, we plot the gIoU curve with respect to miou_{th}. Fig. 10 further shows that

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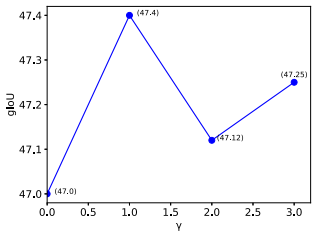


Fig. 9. Curve of gIoU as the value of γ changes.

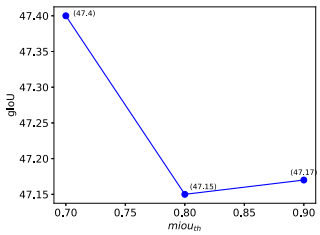


Fig. 10. Curve of gIoU as the value of miou_{th} changes.

the gIoU curve peaks at $\text{miou}_{\text{th}} = 0.7$. Therefore, we set the optimal threshold to 0.7 for the loss design.

C. Comparision Results

1) Comparision with State-of-the-Art Models: Our model is the first method to address the GRRSIS task. To evaluate its effectiveness, we compare its performance with three categories of current state-of-the-art models on the GRRSIS-D dataset: 1) GRES approaches designed for natural images, such as the ReLA [15] and MABP [47] model; 2) RRSIS approaches designed for remote sensing data, including RMSIN [8] and LGCE [7]; and 3) RES approaches designed for natural images, such as CSMA [42], LSCM [43], BRINet [2], CMPC+ [44], ReSTR [4], LAVT [30], CrossVLT [45], and CMIRNet [46]. Since the methods in (2) and (3) are not designed to handle no-target cases, they lack the ability to distinguish between positive and negative samples. In this comparison, we considered samples with fewer than 20 positive pixels in the predicted mask as being classified by the model as no-target samples. To ensure fairness in the comparison, the parameters of the state-of-the-art models involved in the comparison are kept consistent with the code released in the original article.

The comparison results are shown in Tables VIII and IX. Table VIII presents the performance across the entire dataset, while Table IX shows results on target samples only. In these tables, the best results are marked in bold, and the second-best results are underlined. As indicated in both tables, methods in categories 2) and 3) perform poorly on the GRRSIS-D dataset. Specifically, although these models typically achieve high N-acc values (above 85%), indicating strong recognition accuracy for no-target samples, their T-acc values are generally below 60%. The lower T-acc indicates that the presence of no-target samples impairs these models' capacity to accurately segment target objects. In other words, a large number of target samples tend to be misclassified as no-target samples. This limitation stems from the fact that these methods are not designed to handle no-target scenarios, highlighting the necessity of explicitly modeling such cases in the GRRSIS task. Compared with categories 2) and 3), our model significantly improves overall performance by reducing misclassifications on target samples and enhancing object segmentation. Specifically, our model achieves notable gains in gIoU, cIoU, P@0.5, P@0.6, P@0.7, and T-acc, while maintaining competitive performance on P@0.8 and P@0.9, despite a slight decline in N-acc. Methods in category 1) also perform suboptimally on GRRSIS-D. This is likely due to their lack of adaptation to the unique challenges of remote sensing imagery, such as variations in object orientation. Compared with these approaches, our model exhibits superior performance, particularly in accurately segmenting target samples and identifying no-target samples. Overall, our model achieves state-of-the-art performance on the GRRSIS-D dataset, demonstrating its robustness and effectiveness in both no-target and target scenarios.

TABLE X
MODEL COMPLEXITY ANALYSIS

	GFLOPS	Params (M)	Memory (MB)	Infer (ms)
CMSA [42]	220.81	81.41	310.57	55.00
LSCM [43]	268.09	128.56	490.41	56.52
BRINet [2]	742.80	239.27	912.74	103.27
CMPC+ [44]	260.52	119.09	454.32	52.00
ReSTR [4]	126.59	126.16	481.24	40.47
LAVT [30]	197.10	118.85	453.37	45.65
CrossVLT [45]	206.83	236.71	902.20	63.41
CMIRNet [46]	271.16	191.47	730.00	63.39
RMSIN [8]	205.62	261.28	931.08	71.09
LGCE [7]	206.99	167.37	638.46	52.13
ReLA [15]	131.00	225.52	860.30	81.75
MABP [47]	128.78	229.12	874.02	86.58
Ours	150.82	244.24	931.69	83.45

2) Qualitative Results: The visualization comparison of model results clearly demonstrates the advantages of our model over others. The visualization results of all models from categories 1), 2), and 3) are shown in Fig. 11. For each sample, the following are displayed in sequence: ground truth, results of the CMSA model, LSCM, BRINet, CMPC+, ReSTR, LAVT, CrossVLT, CMIRNet, RMSIN, LGCE, ReLA, MABP, and our model. Compared with the other models, our model exhibits a superior ability to distinguish between target and no-target samples and accurately outputs clear segmentation boundaries.

3) Model Complexity: Model complexity is indeed an important metric for evaluating a model. To provide a more precise assessment of its practical deployment, we report the giga floating-point operations per second (GFLOPs), model size, memory consumption, and inference time (measured on an NVIDIA RTX 2080Ti GPU) for our OMSN model and compare these metrics with the three aforementioned types of models. The results are presented in Table X.

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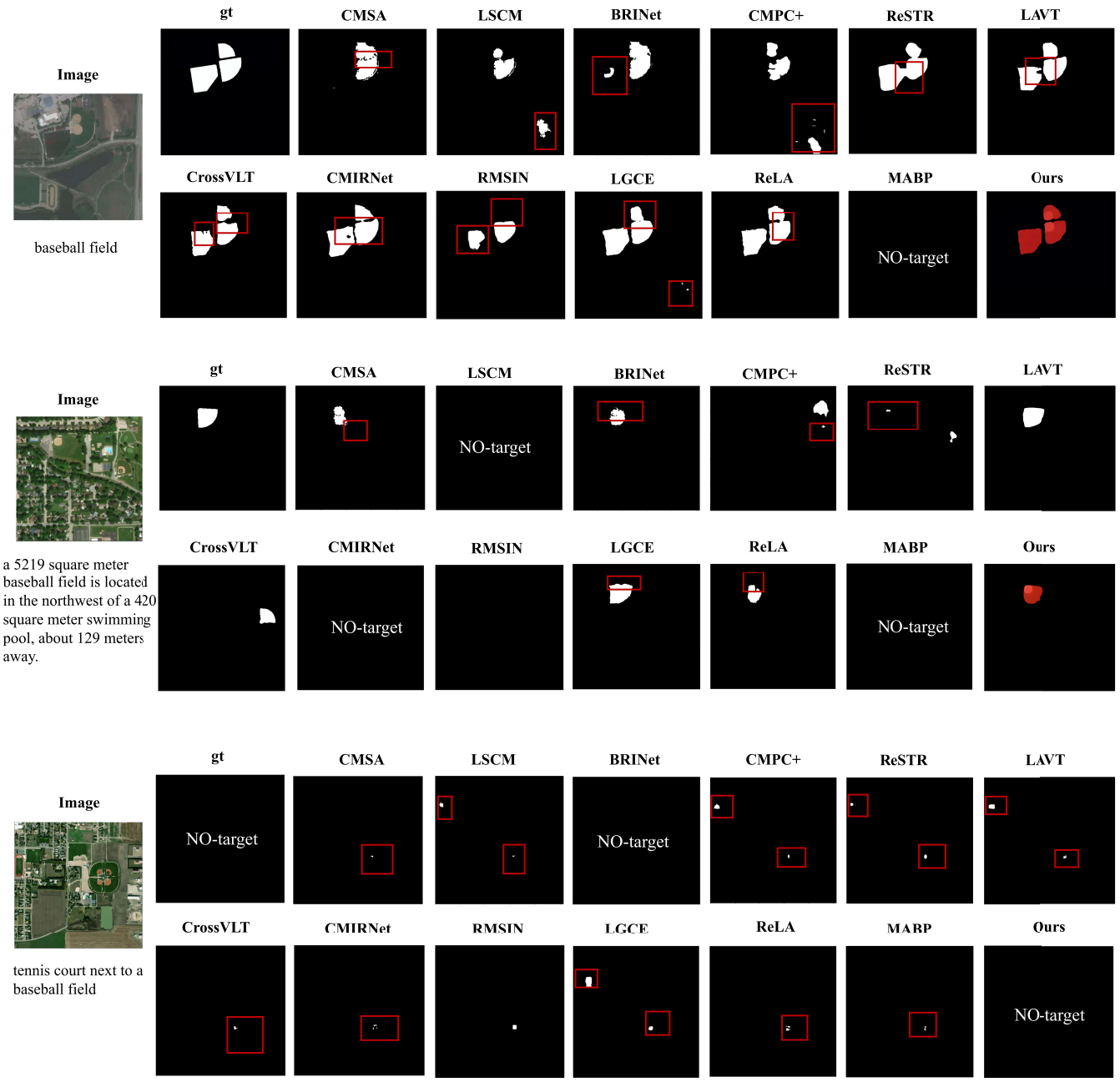


Fig. 11. Visual comparison between our method and state-of-the-art approaches on representative GRRSIS-D samples. Critical regions are highlighted with red boxes for focused analysis.

As shown in Table X, OMSN (Ours) achieves a balance between model capacity and efficiency. Although it has a slightly larger model size (244.24M) compared with most other models, this contributes to its enhanced feature representation capability, leading to superior performance. In addition, OMSN achieves an inference time of 83.45 ms per image on an NVIDIA RTX 2080Ti GPU, which is comparable to that of existing models.

4) Results on Classic RRSIS: Moreover, our model can also be applied to the classic RRSIS task. However, since all training data consist of target samples, the classification branch is no longer utilized. To further validate the effectiveness of our model, we tested it on the classic RRSIS task. Table XI presents a comparison between our model and current state-of-the-art methods on two classical datasets, i.e., RefregRS and RRSIS-D. Clearly, our model achieves performance comparable to existing state-of-the-art models in the classic RES task. From Table XI, we observe that while our model generally outperforms the ReLA model on both datasets, its accuracy at P@0.8–P@0.9 is lower than that of ReLA. This is due to our setting of miou_{th} to 0.7 during training, which reduces the emphasis on P@0.8–P@0.9 samples. In practical applications, miou_{th} can be adjusted based on specific requirements.

D. Discussion

1) Multimodal Large Models for GRRSIS: Currently, general-purpose vision-language large models (VLMs) have

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TABLE XI
RESULTS COMPARISON ON REFREGRS AND RRSIS-D DATASET

Methods	RefregRS							RRSIS-D						
	P@0.5	P@0.6	P@0.7	P@0.8	P@0.9	gIoU	cIoU	P@0.5	P@0.6	P@0.7	P@0.8	P@0.9	gIoU	cIoU
CSMA [42]	28.07	20.25	12.71	5.61	0.83	41.47	64.53	55.32	46.45	37.43	25.39	8.15	48.54	69.39
LSCM [43]	45.73	27.24	13.86	3.6	0.66	45.56	64.6	56.02	46.25	37.70	25.28	8.27	49.92	69.05
BRINet [2]	22.56	15.74	9.85	3.52	0.56	32.87	60.16	56.90	48.77	39.12	27.03	8.73	49.65	69.88
CMPC+ [44]	34.83	22.51	13.59	3.96	0.49	40.86	63.83	57.65	47.51	36.97	24.33	7.78	50.24	68.64
ReSTR [4]	32.53	22.40	13.15	5.83	1.32	41.52	65.41	58.20	50.47	42.29	31.80	14.19	52.99	72.68
LAVT [30]	71.44	57.4	32.14	15.41	4.51	57.74	76.46	69.52	63.63	53.29	41.60	24.94	61.04	77.19
CrossVLT [45]	75.78	63.57	40.45	17.45	5.23	60.94	77.56	69.55	63.26	53.84	42.06	25.51	60.90	76.55
CMIRNet [46]	73.86	66.26	52.39	29.28	8.15	62.09	79.39	69.29	63.14	53.43	42.60	25.17	60.58	76.36
RMSIN [8]	46.06	28.12	14.97	7.65	1.6	43.98	69.80	66.38	59.02	51.09	41.32	23.97	59.18	77.02
LGCE [7]	73.75	61.14	39.46	16.02	5.45	59.96	76.81	65.23	58.05	49.25	39.77	23.05	57.3	74.2
MABP [47]	77.22	69.51	58.23	38.86	13.87	66.72	73.26	58.05	53.19	45.34	36.22	22.48	51.69	65.48
ReLA [15]	80.05	67.98	53.60	**36.43**	**17.87**	66.51	72.85	72.43	66.22	55.61	**44.19**	**25.59**	63.52	**77.77**
Ours	**81.84**	**69.73**	**53.71**	34.56	14.58	**69.26**	**84.39**	**74.68**	**66.97**	**56.16**	44.01	23.94	**65.22**	77.09

TABLE XII
PERFORMANCE OF GENERAL-PURPOSE MULTIMODAL LARGE MODELS ON THE GRRSIS TASK

	gIoU	cIoU	P@0.5	P@0.6	P@0.7	P@0.8	P@0.9	T-acc	N-acc
LLaVA-plus	0.50	0.27	0.24	0.21	0.15	0.03	0	100	0
SEEM	18.28	2.39	0.75	0.42	0.11	0.05	0	55	51
Ours	47.41	42.63	31.97	24.78	16.83	8.71	1.61	83.42	74.45

gained significant attention and become a research hotspot. In this section, we evaluate the performance of two representative VLMs, LLaVA-plus [48] and SEEM [49], on the GRRSIS task. However, when directly feeding our expressions into LLaVA-plus, the model failed to return masks for all samples. To address this, the expressions are reformulated into the form of "detect and segment + original expression." With this modification, only 5086 out of 17 110 test samples produced valid masks, and the results reported in Table XII are based on these correctly processed cases. The overall performance remains far from satisfactory, indicating that LLaVA-plus is not effective for GRRSIS. In contrast, SEEM is able to generate a mask for each sample; however, as shown in Table XII, its segmentation accuracy is still significantly lower than that of our proposed model. These findings demonstrate that, although general-purpose multimodal large models are powerful in natural image scenarios, they are not well-suited for the GRRSIS task.

2) Limitation and Future Work: Despite the remarkable performance demonstrated on various benchmarks, our model still exhibits certain limitations. We discuss these limitations in this section to inspire future research and development.

First, although OMSN is comparable to existing models in terms of model size and runtime, there remains room to further reduce parameter count and improve inference speed. Since GRRSIS is a multimodal, cross-domain task involving feature extraction and fusion across both visual and language modalities, it inherently requires a larger number of parameters and longer inference time. Recently, various model compression techniques—such as knowledge distillation, quantization, and pruning—have attracted increasing attention for their ability to reduce computational overhead without significantly compromising performance. Exploring these techniques to achieve a better tradeoff between computational cost and model accuracy is a promising direction for future research. In addition, replacing Transformers with lightweight and efficient backbones such as MobileFormer and Mamba to simplify the computational structure of the model also represents a viable research avenue.

Second, our model is primarily designed for real-world top-view urban remote sensing scenarios and does not explicitly handle certain special cases, such as occlusions from uncrewed aerial vehicle (UAV) perspectives or highly cluttered environments in specialized remote sensing imagery. These scenarios introduce additional challenges, including severe viewpoint variations and frequent object occlusions, which may affect the model's robustness and generalization ability. Extending the GRRSIS task to address these special scenarios offers a promising direction for future research, particularly to enhance robustness and transferability across diverse remote sensing conditions.

Last, although our proposed online hard example mining loss effectively enhances the model's ability to detect small objects and reduces the miss rate, the continuous downsam-

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pling operations during feature extraction inevitably lead to the loss of semantic features for tiny objects. As a result, some of these objects still cannot be accurately segmented. Therefore, exploring efficient strategies to preserve features of tiny objects represents an important direction for future research.

VII. Conclusion

In this work, we first identify the limitations of the classic RRSIS task and propose a new benchmark, GRRSIS, which alleviates these limitations by allowing expressions to refer to targets that may not be present in the image. To support this task, we introduce GRRSIS-D, a large-scale remote sensing dataset. Considering the unique characteristics of remote sensing imagery, we design a baseline method called OMSN, which proposes an AMFF module based on adaptive rotational convolutions to better accommodate objects with diverse orientations. In addition, we propose an online hard case mining loss to supervise model training and address challenges posed by remote sensing scenes and no-target samples. Our model achieves new state-of-the-art results on both the classic RRSIS task and the newly proposed GRRSIS task. The introduction of GRRSIS significantly relaxes constraints on natural language inputs, thereby broadening its potential applications.

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